



Birla Institute of Technology & Science, Pilani

Hyderabad Campus

FIRST SEMESTER 2026-2027

Course Handout Part II

Date: 01.08.2026

In addition to Part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : **MATH U110**
Course Title : **Essential Precalculus and Calculus**
Instructor-in-charge : **ASHISH TIWARI** (6067-H, ashish.tiwari@pilani.bits-pilani.ac.in)
Instructors : **Rakhee** (6067-R, rakhee@pilani.bits-pilani.ac.in)

1. Course description: Review of coordinate geometry, Rational Fractions, Complex Numbers and quadratic equations, Matrices and determinants, Progression and series, permutations and combinations, Binomial theorem, Functions: Polynomial, Trigonometric, exponential, logarithmic etc., One variable Calculus: Limit and continuity, Differentiation, Integration. Applications of derivatives and definite integration. Program-specific applications of the above mathematical concepts.

2. Scopes and objective of the course:

This Course deals with intermediate mathematics that is essential for PCB students who joined BE (E&S engineering). Course covers set theory, Functions, Coordinate geometry, basic algebra & theory of equations, permutations and combinations, Binomial theorem, Trigonometry, One Dimensional Calculus: Limit and continuity, Differentiation, Integration, Application of derivatives and definite integration.

3. Course learning outcomes:

- CLO1.** Apply concepts of algebra, trigonometry, matrices, and discrete mathematics to solve mathematical problems.
- CLO2.** Analyze and solve problems in coordinate geometry, three-dimensional geometry, and differential calculus.
- CLO3.** Apply integral calculus and related mathematical techniques to solve real-world problems.

4. Text Books and References:

- TB1.** Mathematics for Class XI: Text book for CBSE national council of educational research and training.
- TB2.** Mathematics for Class XII part I: Text book for CBSE national council of educational research and training.
- TB3.** Mathematics for Class XII part II: Text book for CBSE national council of educational research and training
- RB1.** Thomas Finney: Calculus & analytic geometry 14th edition Pearson
- RB2.** Stewart: Calculus early transcendentals 5e 2003 Thomson.



5. Lecture Plan:

Lect No.	Module	Lecture Plan	Article	Learning objectives
1	Theory of sets	L1-Sets, operation on sets, finite and infinite set, power set L2- Cartesian product, relations and functions	Chapter 1 & 2 of text book I	Students will understand the concept of set theory, basics of functions (domain, codomain, range) and relations
2	Trigonometry	L3: Trigonometric Functions and their Identities, L4: Simple Trigonometric Equations, L5: Trigonometric Functions of Sums and Differences of Two Angles, L6: Inverse Trigonometric Functions.	Chapter 3 of text book I, Chapter 2 text book 2	Students will get exposure to the concepts of right triangle trigonometry, understand the definitions of the inverse trigonometric functions. They will be able to know and apply identities involving the trigonometric functions, Finding solutions of a trigonometric equation
3	Complex numbers	L7: Complex numbers, L8: Quadratic equations	Chapter 4 of text book I	Students will get an exposure to the complex number, sum, difference, product and quotient of two complex numbers, conjugate of a complex number, modulus and argument of a complex number, solving quadratic equation.
4	Permutations and combinations	L9: Permutations, L10: Combinations	Chapter 6 of text book 1	Understanding of fundamental principle of counting, deriving relation between permutation and combination.
5	Binomial Theorem	L11-L12: Binomial theorem for positive integer power	Chapter 7 of text book 1	Exposure to proof of Binomial theorem for positive integral values, finding the value of a given number using binomial theorem, finding general and middle terms.
6	Series	L13: Arithmetic progression, geometric progression, Arithmetic mean, geometric mean, L14: Infinite series, infinite geometric series, exponential and logarithmic	Chapter 8 & appendix 1 of text book 1	Students will learn the concept of sequence and series, progression and their types, definitions and properties of progression, discussion on various types of series.



		series		
7	Two dimensional geometry	L15: Condition for parallelism and perpendicularity of two lines, L16: Angle between lines Equations of line in various forms(slope, intercept, through given two points, slope point, general) L17: distance of a point from a line	Chapter 9 of text book 1	Understand the concept of parallel and perpendicular lines, linear equations in various forms, solutions of linear equations, geometric interpretations of equations.
8	Conics	L18: Conic sections, eccentricity, latus rectum, Locus, L19: Circle, parabola, L20: Hyperbola, ellipse, pair of lines.	Chapter 10 of text book 1	Understanding how circle, ellipse, parabola and hyperbola form the sections of a cone, finding standard equation of circle, parabolas, ellipses and hyperbolas and properties of each of them.
9	Three Dimensional Geometry	L21-L22: Three dimensional geometry (distance, equations of line and plane in space, distance of a point from plane)	Chap. 11 of text book 1 & Chap. 11 of text book 3	Exposure to concept of coordinate axes and coordinate planes in three dimensional spaces, finding coordinates of a point in space, compute the distance between two points, section formula and midpoint of a line segment.
10	Matrices	L23-L24: Matrices and determinants; basic properties	Chapter 3 of Textbook 2.	Students will understand the basic matrix theory.
11	Differential calculus	L25-L28: Limits, continuity, differentiability, higher order derivatives, L29-L31: Chain rule Logarithmic differentiation, mean value theorem, Rolle's theorem, L32-L34: Slope of tangents, maxima and minima, indeterminate forms. Program specific	Chap 12 textbook 1. Chap. 5 & 6 text book 2	Students will be able to learn meaning and concept of limits and continuity, understanding differentiation of a function, various theorems based on limit, continuity and differentiability, applications in various forms, concept of maxima and minima.



		applications		
12	Integral Calculus	L35-L37: Anti-derivatives and indefinite integrals, Methods of substitution, parts, partial fractions, trigonometric reduction formulas, L38-L40: Fundamental theorem of calculus, Definite integrals, area under curve. Program specific applications	Chapter 7 & 8 of text book 3	Understanding of integration, methods of integrations, understanding fundamental theorem of calculus, Definite integral and its usage in finding area under the curves.

6. Evaluation Scheme:

EC No.	Evaluation Component	Duration	Weightage (%)	Date & Time	Nature of Component
1.	Quizzes/Class Performance Tests	20 min	20	To be Announced	Closed Book
2.	Mid semester Exam	90 min	35	06/10 AN2	Closed Book
4.	Comprehensive Exam	180 min	45	07/12 AN	Closed Book/Open Book

6. Announcements: All announcements in relation to the above course will be put up on course page in LMS.

7. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

8. Make up policy: Make up for the mid-semester/comprehensive examination will be given to genuine cases.

8. Chamber consultation hours: TBA

9. Academic Honesty and Integrity Policy: Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

10. Criterion for NC: Below the 10% marks or 30% of the median whichever is lower.

Instructor In-Charge

MATH U110

